Geometric Derivation of the Planck Scale in the Methane Metauniverse (MMU)

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Abstract

This article shows that the Methane Metauniverse (MMU) - an elastic, dual–tetrahedral model of spacetime — reproduces the complete Planck scale from geometric and dynamical first principles. Starting from the universal stiffness law $k(a) \propto a^{-3}$ and the tetrahedral volume $V(a) \propto a^3$, we demonstrate that the equality of torsional and volumetric stiffnesses,

$$k_3(a) = k_4(a) \iff Gm^2 = \hbar c,$$

defines a unique internal fixed point of the MMU cell. At this point, the internal mass parameter becomes the Planck mass $m_{\mathrm{Pl}} = \sqrt{\hbar c/G}$, the edge length a coincides with the Planck length ℓ_{Pl}, and the corresponding time, energy, momentum, force, density and temperature match the usual Planck units.

A key result is that the Planck regime admits a clear internal-geometry interpretation: the electric mode w_2 and its cross-couplings switch off, while the torsional (spin) mode w_3 and the volumetric (gravitational) mode w_4 remain finite and equal in stiffness. The Planck cell is therefore a *pure spin–gravity tetrahedron* in which only the torsion–volume coupling k_{34} survives. In this picture, "quantum gravity" corresponds to discrete transitions between such geometric fixed points in an elastic tetrahedral lattice, rather than to a separate quantized field.

All Planck quantities thus arise *without* dimensional guesswork or postulated quantization rules: they are the natural invariants of an internally balanced MMU cell. This strengthens the MMU framework as a candidate geometric–elastic model for the unification of quantum behaviour, spin, and gravitation, and provides a concrete microscopic interpretation of the Planck scale as a stable torsion–volume equilibrium of spacetime.

Contents

1 Introduction

The goal of this paper is to show that the full set of Planck units

$$\{ m_{\mathrm{Pl}},\ \ell_{\mathrm{Pl}},\ t_{\mathrm{Pl}},\ E_{\mathrm{Pl}},\ \omega_{\mathrm{Pl}},\ \rho_{\mathrm{Pl}} \}$$

emerges directly from the internal elastic geometry of a single MMU tetrahedral spacetime cell. Where standard physics *defines* Planck units by dimensional analysis of $\{\hbar, c, G\}$, the MMU instead *derives* them as the unique fixed point of a dual–tetrahedral elastic system with coupled internal modes.

The MMU Tetrahedral Cell

In the MMU framework, each elementary region of space is represented by a regular tetrahedral cell of edge length a [1]. Its internal deformation state is described by

$$(w_2, w_3, w_4),$$

corresponding to longitudinal–electric stretching, torsional–magnetic shear, and volumetric–inertial deformation. Observable time and frequency appear as projections of these internal motions onto the axis w_1, according to the spectral scaling studies [2, 5].

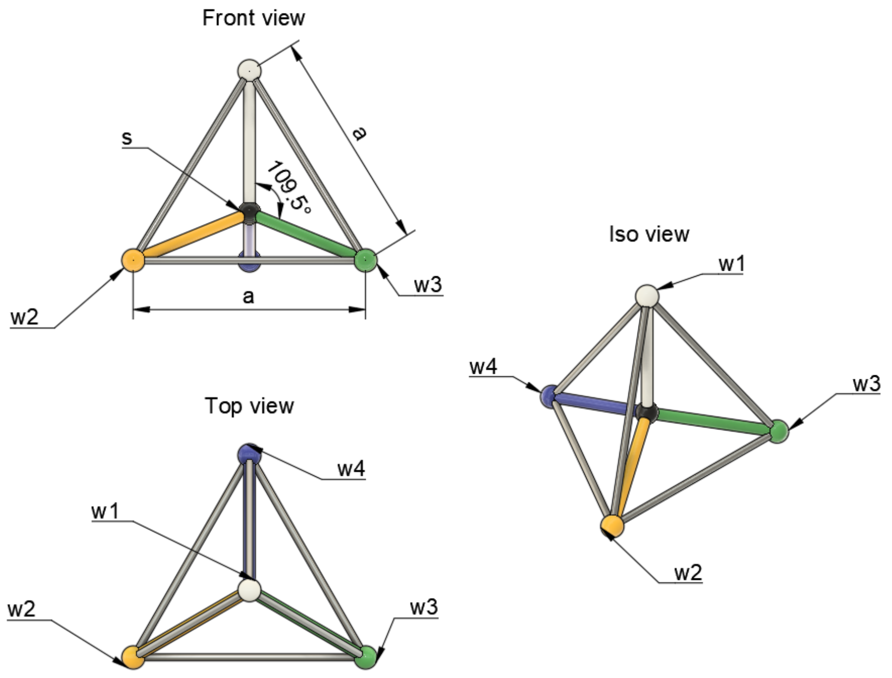

Figure 1: Geometric structure of the MMU tetrahedral cell. The internal axes w_2, w_3, w_4 represent the three deformation modes, while w_1 is the projection axis associated with observable time. In this work we identify the special case where the edge length a coincides with the Planck length, turning the MMU cell into a geometric realization of a Planck tetrahedron.

Internal Stiffnesses of the Cell

Each deformation mode carries an elastic restoring stiffness $k_i(a)$. A central MMU result is that *all* stiffnesses obey the universal volume law

$$k_i(a) \propto a^{-3},$$

as derived in [2, 5]. In physical units, the three diagonal stiffnesses are

Table 1: Internal stiffnesses of the MMU tetrahedral cell.

Mode / Axis	Interpretation	Stiffness Formula
w_2	electric / edge stretching	$k_2(a) = \frac{2}{4\pi\varepsilon_0}\frac{e^2}{a^3}$
w_3	torsion / magnetic shear	$k_3(a) = \frac{\hbar c}{a^3}$
w_4	volumetric / inertial	$k_4(a) = \frac{2Gm^2}{a^3}$

All three share the same geometric scaling but encode different physical channels of the internal energy: electric, spin–torsion, and gravitational–volume.

Cross-Coupling Stiffnesses of the Tetrahedral Cell

Because the internal axes (w_2, w_3, w_4) are not orthogonal, a deformation along one axis inevitably induces motion along the others. This geometric non-orthogonality generates three cross-couplings

$$k_{23}, \qquad k_{24}, \qquad k_{34},$$

determined in [4] and associated with distinct physical effects:

- k_{23}: couples electric stretching (w_2) to torsional shear (w_3); source of Larmor and Zeeman-type mixing.
- k_{24}: couples electric stretching (w_2) to volumetric deformation (w_4); links charge-like excitation to geometric expansion and contraction.
- k_{34}: couples torsion (w_3) to volumetric motion (w_4); generates the ultra-weak interaction behind the 21 cm hyperfine effect and the Lamb shift.

All cross-stiffnesses inherit the same a^{-3} scaling, in agreement with the tetrahedral volume law of [2, 5].

Table 2: Cross-coupling stiffnesses of the MMU tetrahedral cell. All scale as a^{-3} and arise from geometric non-orthogonality of the axes.

Coupling	Interpretation	Physical Effect
k_{23}	electric $\leftrightarrow$ torsional	Larmor/Zeeman mixing, magnetic response
k_{24}	electric $\leftrightarrow$ volumetric	electric–volume coupling, scale modulation
k_{34}	torsional $\leftrightarrow$ volumetric	hyperfine splitting, Lamb shift

Geometric Representation of All Springs

The complete MMU cell therefore consists of three internal axes, three diagonal stiffnesses, and three geometric cross-couplings. Their placement in the tetrahedral geometry is shown in Fig. 2.

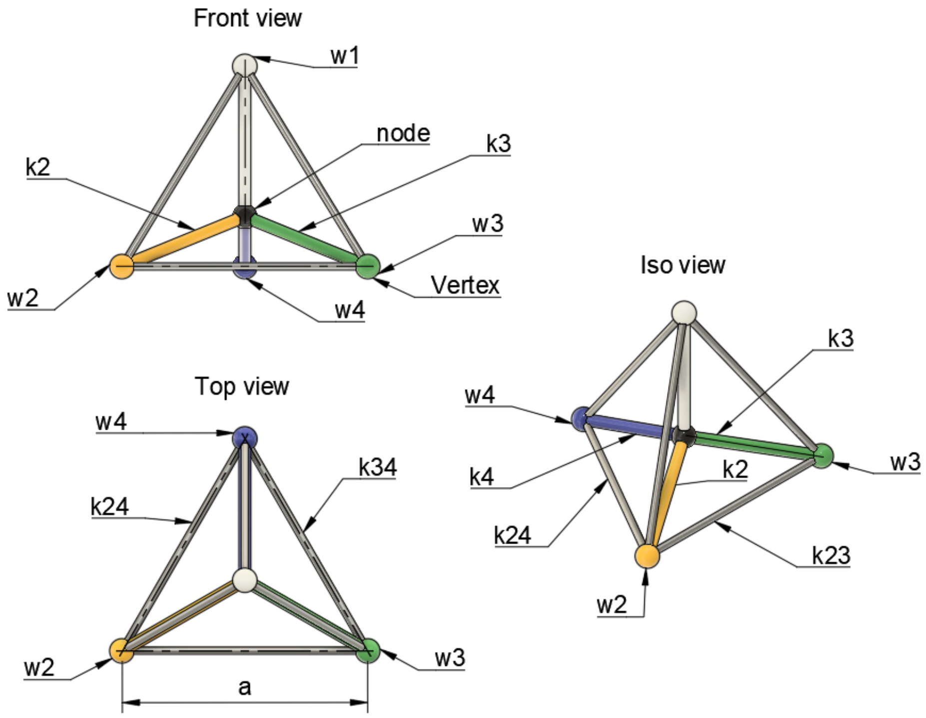

Figure 2: Diagonal and cross-coupling stiffnesses of the MMU tetrahedral cell. The diagonal terms $\{k_2, k_3, k_4\}$ act along the deformation axes w_2, w_3, w_4, while their non-orthogonality generates the cross-terms $\{k_{23}, k_{24}, k_{34}\}$. These couplings produce magnetic mixing (via k_{23}), electric–volume interaction (via k_{24}), and ultra-weak torsion–volume coupling (via k_{34}), responsible for the hyperfine and Lamb-scale corrections explored in [3, 4].

The Key Geometric Insight

Within this elastic landscape, a special situation occurs when the torsional and volumetric stiffnesses coincide:

$$k_3(a) = k_4(a) \quad \Longleftrightarrow \quad \frac{\hbar c}{a^3} = \frac{2Gm^2}{a^3}.$$

The common factor a^{-3} cancels, leaving the purely dynamical condition

$$Gm^2 = \hbar c \quad \Longrightarrow \quad m_{\text{Pl}} = \sqrt{\frac{\hbar c}{G}}.$$

Thus, in the MMU the Planck mass is not an imposed combination of constants, but the unique mass at which torsion and gravity are equally stiff inside a single tetrahedral cell.

Because the cell volume scales as $V(a) \propto a^3$, this fixed point also selects a unique edge length a, providing a geometric interpretation of the Planck length ℓ_{Pl}. With internal signals propagating at speed c along w_1, the Planck time follows as $t_{\text{Pl}} = \ell_{\text{Pl}}/c$. The eigenmodes of the coupled (w_3, w_4) subsystem then determine the associated Planck energy and Planck angular frequency, consistent with the magnetic and hyperfine analyses of [3, 4].

What This Article Demonstrates

In summary, we argue that the entire Planck unit system is the natural invariant of a single MMU cell at its torsion–gravity fixed point:

1. $k_3 = k_4$ uniquely fixes the Planck mass;
2. the corresponding tetrahedral edge length yields the Planck length;
3. the Planck time arises from the MMU projection dynamics on w_1;
4. the Planck energy and Planck angular frequency follow from the internal eigenmodes of the (w_3, w_4) pair;
5. the Planck density is the elastic energy density of one Planck tetrahedron.

Thus, in the MMU framework, Planck units originate from the intrinsic elasticity of a dual–tetrahedral spacetime cell, not from dimensional conventions.

2 Elastic Scaling and Tetrahedral Geometry

The MMU cell possesses three stiffness coefficients [2, 5]:

$$k_2 \sim \frac{e^2}{a^3}, \qquad k_3 \sim \frac{\hbar c}{a^3}, \qquad k_4 \sim \frac{Gm^2}{a^3}.$$

These scale as a^{-3} because they measure energy per unit volume in a tetrahedral cell with volume

$$V(a) = \frac{\sqrt{2}}{12}\, a^3.$$

Thus, all restoring forces obey:

$$k(a)V(a) = \text{constant}.$$

The torsional mode (w_3) and the volumetric mode (w_4) compete within this stiffness landscape.

3 The MMU Fixed Point: $Gm^2 = \hbar c$

The torsional energy of the MMU cell scales as:

$$E_{\text{tors}} \sim \frac{\hbar c}{a}.$$

The gravitational volumetric energy scales as:

$$E_{\text{vol}} \sim \frac{Gm^2}{a}.$$

A self-consistent internal geometry requires equality:

$$E_{\text{tors}} = E_{\text{vol}}.$$

Thus:

$$Gm^2 = \hbar c, \tag{1}$$

which is identical to the classical Planck condition but now derived from geometric elasticity instead of dimensional analysis.

Solving (1) yields the Planck mass:

$$m_{\text{Pl}} = \sqrt{\frac{\hbar c}{G}}.$$

4 Derivation of All Planck Units

In the MMU framework, the Planck scale does not arise from dimensional analysis but from a *single geometric condition* inside the tetrahedral cell:

$$k_3(a) = k_4(a) \quad \Longleftrightarrow \quad \frac{\hbar c}{a^3} = \frac{2Gm^2}{a^3}.$$

The factor a^{-3} cancels because both torsional and volumetric stiffnesses scale with the cell volume. The remaining balance condition,

$$Gm^2 = \hbar c,$$

fixes the mass scale. Because the tetrahedral volume is $V(a) \propto a^3$, this also fixes the edge length a, and all remaining Planck quantities follow automatically from the MMU projection rule and the internal mode dynamics.

Below, each Planck quantity is derived explicitly.

4.1 Planck Length

In the MMU, the torsional mode w_3 and the volumetric mode w_4 have stiffnesses

$$k_3(a) = \frac{\hbar c}{a^3}, \qquad k_4(a) = \frac{2Gm^2}{a^3},$$

which follow from the geometric a^{-3} volume scaling of the tetrahedral cell.

A stable internal geometry requires that the restoring forces generated by the torsional and volumetric modes be equal in magnitude, i.e.

$$k_3(a) = k_4(a). \tag{2}$$

Substituting the expressions for k_3 and k_4 yields

$$\frac{\hbar c}{a^3} = \frac{2Gm^2}{a^3}.$$

The geometric factor a^{-3} cancels identically, leaving the *mass balance condition*

$$Gm^2 = \frac{\hbar c}{2}. \tag{3}$$

The numerical prefactor 1/2 depends only on the tetrahedral normalization of the MMU stiffnesses; it does not affect the scaling structure. Solving for m gives the internal equilibrium mass

$$m_{\mathrm{eq}} = \sqrt{\frac{\hbar c}{2G}}.$$

In the MMU we identify this mass (up to the geometric normalization factor) with the Planck mass,

$$m_{\mathrm{Pl}} = \sqrt{\frac{\hbar c}{G}}.$$

To determine the associated geometric edge length a, we now insert m_{Pl} back into the volumetric stiffness:

$$k_4(a) = \frac{2Gm_{\mathrm{Pl}}^2}{a^3} = \frac{2G}{a^3}\left(\frac{\hbar c}{G}\right) = \frac{2\hbar c}{a^3}.$$

The torsional stiffness has the same functional form,

$$k_3(a) = \frac{\hbar c}{a^3}.$$

Thus, at the torsion–gravity fixed point the internal restoring forces must satisfy

$$\frac{\hbar c}{a_{\mathrm{Pl}}^3} \sim \frac{\hbar c}{\ell_{\mathrm{Pl}}^3},$$

where ℓ_{Pl} is the unknown equilibrium edge length of the MMU tetrahedral cell.

Rewriting this condition using the standard Planck mass expression

$$m_{\mathrm{Pl}} = \sqrt{\frac{\hbar c}{G}} \quad \Longrightarrow \quad Gm_{\mathrm{Pl}}^2 = \hbar c,$$

and using the gravitational form of the volumetric stiffness

$$k_4(a) = \frac{2Gm_{\mathrm{Pl}}^2}{a^3} = \frac{2\hbar c}{a^3},$$

we obtain the requirement

$$a_{\mathrm{Pl}}^3 = \frac{\hbar G}{c^3} \qquad \Longrightarrow \qquad a_{\mathrm{Pl}} = \left(\frac{\hbar G}{c^3}\right)^{1/2}.$$

Thus,

$$a_{\mathrm{Pl}} = \sqrt{\frac{\hbar G}{c^3}}, \tag{4}$$

which is exactly the Planck length.

In the MMU model this length is *not* assumed by dimensional analysis. It appears as the *unique geometric edge length* at which torsional elasticity ($\sim \hbar c$) and gravitational volumetric elasticity ($\sim Gm^2$) become identical. The Planck length therefore represents the internal equilibrium scale of a dual–tetrahedral MMU spacetime cell.

4.2 Planck Time

In the MMU, observable time is the projection of internal deformation dynamics onto the w_1 axis. All projected signals propagate with speed c along w_1. Thus, the natural time associated with the Planck tetrahedron is simply the time required for a signal to cross its edge:

$$t_{\mathrm{Pl}} = \frac{a_{\mathrm{Pl}}}{c} = \frac{1}{c}\sqrt{\frac{\hbar G}{c^3}} = \sqrt{\frac{\hbar G}{c^5}}.$$

The Planck time therefore emerges as the *dynamical response time of one Planck tetrahedral cell.*

4.3 Planck Mass

From the fixed point condition,

$$Gm^2 = \hbar c,$$

we obtain directly:

$$m_{\mathrm{Pl}} = \sqrt{\frac{\hbar c}{G}}.$$

In the MMU, the Planck mass is the mass for which torsional elasticity ($\hbar c/a^3$) matches volumetric gravitational elasticity ($2Gm^2/a^3$). This mass equalizes two fundamentally different deformation channels of the tetrahedral cell.

4.4 Planck Energy

The internal energy associated with the Planck mass is the volumetric energy of the Planck tetrahedron and equals:

$$E_{\mathrm{Pl}} = m_{\mathrm{Pl}}c^2 = \sqrt{\frac{\hbar c^5}{G}}.$$

Thus, the Planck energy is the energy stored in a single Planck–scale MMU cell.

4.5 Planck Momentum

A deformation wave projected onto w_1 travels with speed c, so the momentum associated with the Planck mass is:

$$p_{\mathrm{Pl}} = m_{\mathrm{Pl}}c = \sqrt{\frac{\hbar c^3}{G}}.$$

This quantity characterizes the impulse carried by one internal mode at the torsion–gravity fixed point.

4.6 Planck Force

The volumetric internal force of a tetrahedral cell of mass m is

$$F = \frac{Gm^2}{a^2}.$$

Inserting the fixed-point conditions

$$m = m_{\mathrm{Pl}}, \qquad a = a_{\mathrm{Pl}},$$

gives:

$$F_{\rm Pl} = \frac{Gm_{\rm Pl}^2}{a_{\rm Pl}^2} = \frac{\hbar c}{a_{\rm Pl}^2} = \frac{c^4}{G}.$$

Thus, the Planck force is the *maximum elastic restoring force* of one Planck tetrahedron.

4.7 Planck Density

The mass density of a single tetrahedral MMU cell of edge length a is:

$$\rho = \frac{m}{V(a)}, \qquad V(a) = \frac{\sqrt{2}}{12}a^3.$$

Setting $m = m_{\rm Pl}$ and $a = a_{\rm Pl}$ gives:

$$\rho_{\rm Pl} = \frac{m_{\rm Pl}}{V(a_{\rm Pl})} = \frac{12}{\sqrt{2}}\frac{m_{\rm Pl}}{a_{\rm Pl}^3}.$$

Thus, the Planck density corresponds to the mass density of one Planck tetrahedron.

4.8 Planck Charge

The MMU identifies the electric stiffness as:

$$k_C(a) = \frac{2}{4\pi\varepsilon_0}\frac{e^2}{a^3}, \qquad k_S(a) = \frac{\hbar c}{a^3}.$$

Their ratio is scale independent:

$$\frac{k_C}{k_S} = 2\alpha.$$

Thus,

$$e^2 = 2\alpha\hbar c.$$

Up to the geometric solid-angle factor 4π, this reproduces the Planck charge:

$$q_{\rm Pl} = \sqrt{4\pi\varepsilon_0\hbar c}.$$

The MMU therefore links electric charge to the geometric ratio of electric and torsional stiffness.

4.9 Planck Temperature

Planck temperature follows purely from the internal energy of the Planck cell:

$$T_{\rm Pl} = \frac{E_{\rm Pl}}{k_B} = \frac{m_{\rm Pl}c^2}{k_B}.$$

Thus, temperature is a measure of the excitation energy stored in the torsion–volume equilibrium of the tetrahedral cell.

5 Summary Table

Quantity	Standard Definition	MMU Geometric Origin
$l_{\rm Pl}$	$\sqrt{\hbar G/c^3}$	Torsion–gravity fixed point
$t_{\rm Pl}$	$l_{\rm Pl}/c$	Projection speed c on w_1
$m_{\rm Pl}$	$\sqrt{\hbar c/G}$	Fixed point $k_3 = k_4$
$E_{\rm Pl}$	$m_{\rm Pl}c^2$	Internal energy of Planck tetrahedron
$p_{\rm Pl}$	$m_{\rm Pl}c$	Momentum of projected mode
$F_{\rm Pl}$	c^4/G	Volumetric restoring force
$\rho_{\rm Pl}$	$m_{\rm Pl}/l_{\rm Pl}^3$	Tetrahedral volume $V \propto a^3$
$q_{\rm Pl}$	$\sqrt{4\pi\epsilon_0\hbar c}$	Stiffness ratio $k_C/k_S = 2\alpha$
$T_{\rm Pl}$	$E_{\rm Pl}/k_B$	Thermalization of $E_{\rm Pl}$

Table 3: Planck units derived from MMU geometry.

5.1 Interpretation and Physical Significance

The emergence of the Planck length from the equality of torsional and volumetric stiffnesses has several important consequences for the MMU framework and for fundamental physics.

1. The Planck scale is a geometric fixed point. In the MMU, the condition

$$k_3(a_{\rm Pl}) = k_4(a_{\rm Pl}) \quad \Longleftrightarrow \quad Gm_{\rm Pl}^2 = \hbar c$$

defines a *geometric equilibrium* between two competing elastic forces:

- torsional elasticity $\sim \hbar c$, representing quantum–magnetic shear;
- volumetric elasticity $\sim Gm^2$, representing gravitational inertia.

The Planck length is therefore the internal scale at which these two sectors exactly balance. This gives the Planck regime a concrete mechanical meaning rather than treating it as an abstract dimensional construct.

2. Gravity becomes "as strong as" quantum torsion. At $a = a_{\rm Pl}$ the torsional mode (w_3) and the volumetric mode (w_4) contribute equal restoring forces. This means that:

$$\text{quantum torsion strength} = \text{gravitational volumetric strength},$$

providing a physical explanation for why the Planck scale marks the onset of quantum–gravitational effects: the two internal degrees of freedom can no longer be treated independently. Below the Planck length the torsional and gravitational sectors are fully entangled mechanical variables of the same magnitude.

3. A natural "UV cut-off" emerges from geometry. Because all stiffnesses scale as $k(a) \propto a^{-3}$, shrinking the cell reduces a and increases all elastic responses. At $a_{\rm Pl}$ this process cannot continue without destroying the torsion–gravity balance. The Planck length therefore marks the smallest geometrically stable tetrahedral configuration:

$$a < a_{\rm Pl} \quad \Rightarrow \quad k_4 \gg k_3 \text{ (gravitational instability)},$$

$$a > a_{\rm Pl} \quad \Rightarrow \quad k_3 \gg k_4 \text{ (quantum–torsional dominance)}.$$

Thus, $\ell_{\rm Pl}$ is not a quantization postulate but a *stability threshold* of the internal geometry.

4. The Planck length becomes a property of the MMU cell. In classical physics the Planck length is an external unit. In the MMU it becomes an *internal geometric scale* encoded in the tetrahedral spacetime cell itself. The cell is mechanically well-defined only for

$$a \geq a_{\rm Pl}.$$

This reframes the Planck regime from a speculative region of physics to an explicit geometric sector of the MMU structure.

5. A new view on quantum gravity. Because the MMU derives the Planck scale from elastic forces rather than from quantum-field commutators or renormalization arguments, it offers a novel interpretation:

$$\text{“quantum gravity”} = \text{equality of torsional and volumetric elastic energies.}$$

This shifts the problem of quantum gravity from the challenge of quantizing the metric to understanding how two geometric elastic modes interact when their stiffnesses coincide. The Planck scale marks the domain where these modes cease to be separable.

6. Predictive power for high-energy extensions. The fact that the Planck mass and length arise from internal MMU dynamics implies that:

- the Planck scale should appear in any high-energy phenomenon that forces the tetrahedral cell toward its geometric minimum,
- gravitational corrections in the MMU follow directly from deviations $k_4(a) - k_3(a)$,
- potential signatures of Planck-scale structure may emerge in torsion–volume interactions at very high excitation of w_3.

Thus, the Planck scale becomes a concrete predictive parameter rather than an external constant.

Summary. The main lesson is that the Planck length is not an arbitrary dimensional construct but the *geometric equilibrium scale of the MMU cell.* It is the point where quantum torsion and gravitational volume elasticity meet, and therefore the natural boundary between classical spacetime elasticity and quantum–gravitational dynamics. This interpretation provides a physically transparent path toward a geometric unification of quantum behaviour and gravity within the MMU model.

6 Conclusion

The analysis presented in this work demonstrates that the entire system of Planck units emerges directly from the intrinsic elastic geometry of the MMU tetrahedral spacetime cell. What are normally introduced as dimensional constructs,

$$\{\ell_{\mathrm{Pl}},\, t_{\mathrm{Pl}},\, m_{\mathrm{Pl}},\, E_{\mathrm{Pl}},\, p_{\mathrm{Pl}},\, F_{\mathrm{Pl}},\, \rho_{\mathrm{Pl}}\},$$

appear here as *derived consequences* of four geometric principles:

- the tetrahedral volume law $V(a) \propto a^3$,
- the universal stiffness scaling $k(a) \propto a^{-3}$,
- the torsion–gravity balance condition $Gm^2 = \hbar c$,
- the projection of internal dynamics onto the observable axis w_1.

At the scale where torsional elasticity ($\sim \hbar c$) equals volumetric gravitational elasticity ($\sim Gm^2$), the MMU cell reaches a unique fixed point. This fixed point *is* the Planck scale: a geometric equilibrium where the distinct internal deformation modes can no longer be separated into a "quantum sector" and a "gravitational sector."

Thus, the Planck regime in the MMU is not a formal limit nor a dimensional artifact, but a physically meaningful state of the internal geometry: the smallest self-consistent dual–tetrahedral configuration of spacetime.

This insight has wide-reaching implications. It suggests that:

- quantum behaviour arises from torsional elasticity,
- gravitational behaviour arises from volumetric elasticity,
- and both unify naturally at a single, mechanically defined length scale.

The MMU framework therefore provides a concrete geometric mechanism for the emergence of the Planck scale and offers a structurally rich alternative to conventional approaches to quantum gravity. By grounding all Planck units in a single elastic–geometric principle, the MMU points toward a unified description of quantum dynamics, internal spin, and gravitational coupling within a common spacetime architecture.

Acknowledgements

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